Forward Deployed Engineer Take-Home Assessment

Context

You have joined an emerging AI company as a Forward Deployed Engineer.

The leadership team is considering entering the AI software-development-assistant market. Developers already have access to increasingly capable products such as Codex, Claude Code, Cursor, and other coding assistants: the CEO believes there may still be room for a differentiated product; the CFO is skeptical.

Their position is straightforward:

> *"Why should we invest in building another AI coding product when some of the largest technology companies in the world already offer mature alternatives?"*

You have been asked to investigate the opportunity and recommend what the company should do.

Your task

Design the smallest credible product and technical strategy that could justify entering this market.

We are not looking for a clone of an existing coding assistant, nor do we expect a production-ready implementation; we want to understand how you reason when the problem is ambiguous and the technical solution is only one part of the decision.

Your proposal should address four questions

1. Understand the problem

Analyze the current category of AI software-development assistants, and identify the most important limitations, trade-offs, or unmet needs you believe still exist.
You do not need to exhaustively compare every product or feature; instead, identify the few limitations that materially influence your proposed opportunity.
You are free to conclude that the company should **not** build another coding assistant if your analysis leads you there.

2. Define your differentiated product

Assuming you identify a credible opportunity, describe the product you would build, explaining:

- The primary problem you are solving (core value proposition);
- What you would deliberately do differently from existing assistants;
- Which attractive features you would explicitly leave out;
- Why would a user or organization switch from, supplement, or choose your product instead of an established alternative.

Your proposal should be narrow enough that a small engineering team could validate the central hypothesis.

3. Design the system

Provide a technical design for the first credible version (you may include a diagram), so you can discuss the decisions you consider most important, such as:

- Model strategy, context collection and management;
- Repository understanding;
- Tool execution and permissions;
- Agent or workflow orchestration;
- Evaluation, security and privacy;
- Observability, human approval and control;
- Latency, cost and failure handling;
- Enterprise integration, if relevant.

Explicitly identify:

- The largest technical risks and assumptions you are making;
- What you would prototype first;
- What you would measure;
- What evidence would cause you to change or abandon your approach.

A working prototype is a part of the evaluation, but keep it intentionally small and use it only to validate important assumptions.

4. Convince the skeptical CFO

Prepare a short executive recommendation for the CFO, assuming the following question:

> *"Microsoft, OpenAI, Anthropic and well-funded startups are already spending enormous amounts of money on this problem. Why should we believe that we can compete, and why shouldn't we simply buy their products?"*

Respond as if this were a real executive conversation, and you should be prepared to defend your recommendation even if the interviewer challenges its assumptions.

Deliverables

Please spend approximately **2/3 hours** on the exercise. **Do not exceed 4 hours;** submit:

- A concise written proposal/presentation;
- A technical architecture or system-design view;
- A functional prototype or supporting artifact you chose to create.

You may use AI tools as part of your normal workflow, and document your development in a private Github repo that can be audited.Be prepared to explain which tools you used, how you used them, what important decisions remained yours and how you validated generated analysis or code. The quality of the submission is not judged by visual polish or the amount of code produced.

Live discussion

During the seniority interview, you will have approximately:

- 10 minutes to present your recommendation;
- 10–15 minutes for a technical walkthrough;
- The remaining time for questions and a changing business scenario.